SMART INDIA HACKATHON 2026

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- **Problem Statement ID – SIH26189**
- **Problem Statement Title – AI-Powered Criminal Network Analysis System**
- **Theme – Blockchain & Cybersecurity**
- **PS Category – Software**
- **Team ID –**
- **Team Name – TechForge**

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PROPOSED SOLUTION

An AI-powered investigation decision-support system that integrates FIRs, reports, call/network records, transactions and surveillance data into a unified Evidence Knowledge Graph. AI extracts entities, connects related records and provides evidence-backed insights.

DATA EXTRACTED

Call Logs
FIR details
Case number
Judgement order

DATA EXTRACTED

Linked FIR
Vehicle Records
Property records
Arrest warranty

HOW IT ADDRESS THE PROBLEM

Our system integrates fragmented data from multiple sources, automatically discovers hidden cross-record connections, intelligently matches identities across common names, spelling variations and aliases, detects contradictions and missing information in evidence, and reduces manual analysis by helping investigators prioritize the most relevant leads.

INNOVATION & UNIQUENESS

Our system uses uncertainty-aware identity resolution to accurately link identities, tracks unknown entities across cases, detects contradictions and missing evidence, recommends the next best evidence to verify, and ensures evidence integrity through tamper detection and auditable AI recommendations.

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TECHNICAL APPROACH

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1. DATA SOURCES

- **1.1 FIRs / Police Reports** — Structured / Unstructured text, locations, incidents
- **1.2 CDR / Network Records** — Caller details, timestamps, locations, duration
- **1.3 Financial Transactions** — Sender, receiver, amount, timestamp, transaction type
- **1.4 CCTV / Images / Videos** — Visual evidence, metadata, timestamps
- **1.5 Criminal History / Intel** — Prior offenses, known associates, case IDs
- **1.6 Other Sources** — OSINT, public records, documents, events

2. TECH STACK

2.1 Python Core Development	2.2 FastAPI Backend API Framework	2.3 Neo4j Graph Database
2.4 PostgreSQL Relational Database	2.5 spaCy NLP / Entity Extraction	2.6 PyTorch AI / ML Models
2.7 Cytoscape.js Graph Visualization	2.8 OpenCV Image / Video Processing	2.9 Docker Containerization
2.10 JWT / RBAC Auth & Access Control	2.11 Elasticsearch Search & Indexing	2.12 Linux / Nginx Deployment & Web Server

3. CORE MODULES

- **3.1 Entity Extraction** — Extracts Persons, Phones, Vehicles, Accounts, Locations, Organizations, Events, Dates.
- **3.2 Identity Resolution Engine** — Matches entities using fuzzy matching, alias detection and multi-attribute comparison with confidence scoring.
- **3.3 Evidence Knowledge Graph** — Stores entities and relationships with evidence metadata (source, time, location, confidence).
- **3.4 Intelligence & Analytics Engine** — Network analysis, cross-case correlation, spatio-temporal analysis, pattern detection.
- **3.5 Investigation Support Engine** — Contradiction detection, gap detection, Next-Best-Evidence recommendations.
- **3.6 Audit & Integrity Layer (Optional)** — Hashing, chain of custody, and immutable logs for evidence & AI recommendations.

4. PIPELINE

01	02	03	04	05	06	07	08
Data Ingestion	**Preprocessing**	**Entity Extraction**	**Identity Resolution**	**Build Knowledge Graph**	**Analytics & Pattern Detection**	**Insights & Recommendations**	**Investigator Action & Feedback**
Collect data from multiple sources.	OCR, text extraction, cleaning, normalization.	Extract key entities & events using NLP & CV.	Match entities and assign confidence scores.	Create entities, relationships, and evidence links.	Network, temporal & cross-case analysis.	Detect contradictions, gaps and recommend next-best evidence.	Human verification, feedback loop and update.

5. IMPLEMENTATION

- **5.1 Investigator Dashboard** — Overview, cases, entities, network, alerts, analytics.
- **5.2 Network Visualization** — Interactive graph of entities and relationships.
- **5.3 Next-Best-Evidence Recommendation** — AI-suggested next steps with confidence score.
- **5.4 Challenge Portal** — Report issues, suggest improvements, view updates.

Data Flow | AI & Analytics | Knowledge Graph | Secure & Auditable | Human-in-the-Loop

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FEASIBILITY AND VIABILITY

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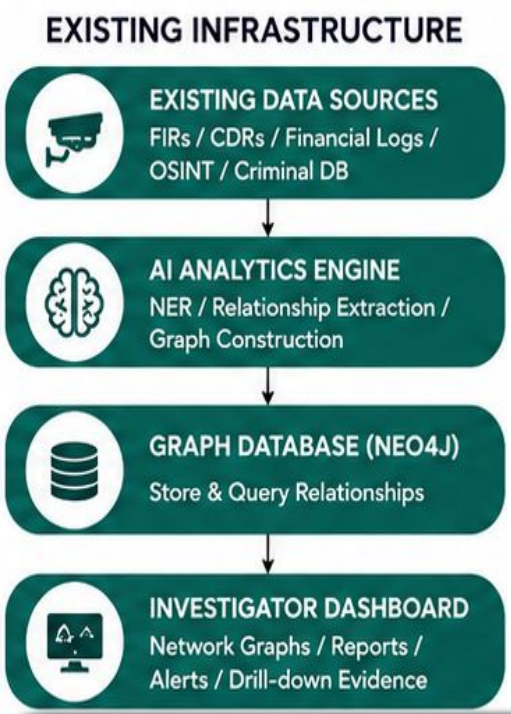
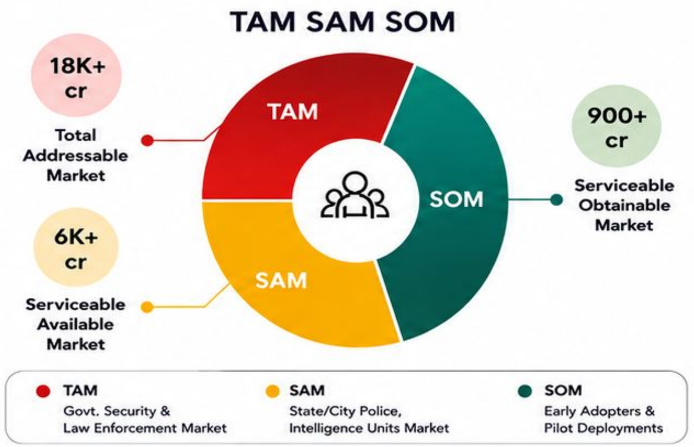
FEASIBILITY

- **Multi-Source Data Support** — FIRs, CDRs, Financial Logs, OSINT, Criminal History
- **AI-Driven Extraction** — NER, Relationship Extraction, Entity Resolution using NLP / LLMs
- **Graph-Based Analysis** — Neo4j enables complex relationship mapping & traversal
- **Real-Time Insights** — Centrality, Anomaly Detection & Risk Scoring
- **Explainable & Verifiable** — Every link traceable to source evidence
- **Scalable Architecture** — Modular & microservices-ready for future scale





- **TAM:** Law enforcement, intelligence, defence and public safety agencies requiring data analytics.
- **SAM:** Police and investigation agencies needing AI-powered criminal network and evidence analysis.
- **SOM:** Initial deployment for selected police units/cybercrime/investigation teams as a pilot.

CHALLENGES FACED

- **Unstructured & Noisy Data**
  - Inconsistent formats in FIRs, reports, OSINT
- **Data Quality & Incompleteness**
  - Missing records, duplicate entities, aliases
- **High Volume & Velocity**
  - Large scale CDRs, transactions, social data
- **Entity Ambiguity**
  - Same person, multiple names/IDs/phones

MITIGATION STRATEGIES

1. Data Processing & Standardization

- Clean and normalize data from FIRs, CDRs, financial records, and OSINT sources to maintain consistent and reliable input formats.

2. Advanced Entity Resolution

- Use fuzzy matching and machine learning models to identify the same person or entity across different datasets.

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IMPACT AND BENEFITS

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BENEFITS

- **Faster Investigation:** Connects fragmented records automatically.
- **Smarter Insights:** Detects hidden links, patterns, and evidence gaps.
- **Better Accuracy:** Handles aliases and name variations.
- **Better Decisions:** Suggests evidence for human verification

Contributes to safer communities through faster identification of criminal networks.

11 SUSTAINABLE CITIES AND COMMUNITIES

Supports effective investigations, crime prevention, and stronger law-enforcement decision-making.

16 PEACE, JUSTICE AND STRONG INSTITUTIONS

SUSTAINABLE DEVELOPMENT GOALS

Can support secure information sharing and collaboration between investigative agencies

17 PARTNERSHIPS FOR THE GOALS

Uses AI and data analytics to modernize investigative infrastructure.

9 INDUSTRY, INNOVATION AND INFRASTRUCTURE

Potential Impact

- Faster investigations by linking fragmented records.
- Evidence-based insights and prioritized leads.
- Reduces manual effort and data overload.

Economic Impact

- Reduces time and manpower required for manual data analysis.
- Improves utilization of existing investigation data and infrastructure.
- Can reduce operational costs by prioritizing high-value investigative actions.

Social Impact

- Enables faster, more effective investigations.
- Supports evidence-based decisions.
- Human verification reduces false conclusions.

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RESEARCH AND REFERENCES

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- Neo4j – Graph Database & Graph Analytics
  https://neo4j.com/
- Neo4j Graph Data Science Library
  https://neo4j.com/docs/graph-data-science/
- NetworkX – Network Analysis in Python
  https://networkx.org/
- spaCy – Natural Language Processing & Entity Extraction
  https://spacy.io/
- FastAPI – Backend API Framework
  https://fastapi.tiangolo.com/
- OpenCV – Computer Vision
  https://opencv.org/
- NIST AI Risk Management Framework (Responsible & Explainable AI)
  https://www.nist.gov/itl/ai-risk-management-framework